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Swin Transformer-Based Temporal-Channel Network for Cross-Subject EEG Emotion Classification

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Abstract

To tackle the challenge of effectively representing time-varying information in cross-subject EEG emotion recognition, we introduce Swin Transformer-Based Temporal-Channel Network (Swin-TCNet), a novel multi-scale neural network with parallel temporal pathways to enhance the extraction of dynamic temporal features. EEG signals are simultaneously processed through a Temporal Swin Transformer for 3D feature representation and a dynamic spatiotemporal convolutional layer with multi-head attention for extracting differential entropy-based channel features, which enhances channel learning while preserving temporal information. Swin-TCNet attains state-of-the-art performance, achieving 93.47% and 86.80% accuracy in cross-subject experiments on SEED and SEED-IV datasets, respectively, as substantiated by ablation studies. By leveraging temporal dynamics, this framework enhances the extraction of temporal variations and spatial information, leading to more robust and generalizable cross-subject emotion recognition.

1. Introduction

Emotion recognition plays a vital role in healthcare and human-computer interaction, offering significant potential to improve experiences in mental health monitoring, market analysis, education, and training [14, 24]. With the advancements in artificial intelligence, emotion recognition models have significantly improved the understanding and interpretation of human emotions. These models based on EEG provide essential support for individuals who face challenges in expressing their feelings [6, 9]. EEG stands out for its ability to directly capture brain activity, making it especially effective in applications that demand authentic and precise emotion detection. EEG-based emotion recognition offers a robust solution for advancing emotion recognition across critical domains.

The majority of research has predominantly focused on within-subject emotion recognition [11], which requires extensive data collection—typically ranging from 30 to 60 minutes—to capture subject-specific EEG patterns [13, 36, 37]. However, in practical applications, emotion recognition is often performed across subjects, making cross-subject generalization essential [17, 25]. The significant inter-individual variability in EEG signals presents a major challenge for cross-subject emotion recognition [1]. For example, the SEED dataset shows a noticeable drop in performance when moving from within-subject to cross-subject recognition using a generic classifier [16]. This decline is attributed to individual differences in EEG-based emotional representations [35]. Nevertheless, consistent emotional representations across subjects have been supported by psychological and neuroscience research, with stable neural patterns identified for various emotions, indicating the potential for effective cross-subject algorithms [12, 38].

EEG signals, renowned for their high temporal resolution, have been integrated into deep learning models increasingly, especially within the field of natural language processing [8]. Previous EEG emotion recognition studies primarily focused on frequency-domain analysis within specific time segments, while overlooking intrinsic temporal and spatial dynamics. This limitation reduced sensitivity to non-stationary signals and constrained the understanding of EEG structure, further emphasizing the necessity of spatiotemporal dynamic fusion [31]. The fusion of spatiotemporal information leverages various deep learning models for feature extraction. In spatiotemporal Transformer-based EEG emotion recognition, the Swin Transformer [21] fuses spatial and temporal information using four-dimensional artificial EEG features through its attention mechanism, reducing noise interference. Similarly, Bi-ViTNet [3] extracts spatiotemporal features by integrating spatial and temporal representations through dual branches with linear embed-

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ding and Transformer encoders. Nevertheless, these methods fail to capture the unique spatiotemporal dependencies of EEG signals, which differ significantly from visual or temporal data, limiting their effectiveness in EEG applications.

Expanding on these advancements, recent studies have further refined deep learning techniques to improve classification efficiency and accuracy. some studies have transformed EEG signals into image formats, focusing on feature extraction within a single time segment [21]. However, this approach often overlooks correlations across different time periods. Other studies have introduced channel attention mechanisms to enhance information extraction from specific channels within a given time segment, yet challenges remain in integrating channel information across multiple time periods [3]. While these methods have achieved some success in feature extraction, they still struggle to capture relationships between features over time and extract multi-interval channel information effectively. Thus, a key challenge in this field is how to efficiently integrate these features to improve the overall accuracy of cross-subject emotion recognition.

The proposed Swin-TCNet addresses the challenges of EEG signal processing by effectively capturing time-varying dynamics and integrating channel-specific information through a dual-path architecture:

1) Parallel Bipartite Processing Architecture: The model utilizes a Channel Attention Module (CAM) and a Time-Variant Signal Attention (TSA) Module to independently capture spatial and temporal features, improving EEG signal analysis.

2) Time-Variant Signal Attention: Utilizes dynamic windowing and hierarchical feature extraction to model temporal dependencies, improving temporal feature extraction with relative position encoding.

3) Channel Attention Module: Applies attention mechanisms to highlight important channels, capturing spatial correlations and enhancing the model's ability to focus on key features.

2. RELATED WORKS

2.1. Transformer

To capture long-range dependencies in EEG data, attention-based methods have increasingly been employed to improve emotion recognition performance [22, 28]. As a self-attention-driven deep learning model, the Transformer has gained significant attention [29], achieving state-of-the-art results in NLP and computer vision domains, exemplified by models such as BERT [4] and ViT [5]. The multi-head self-attention mechanism and parallel input structure enable the Transformer to effectively capture these dependencies. Moreover, positional embeddings preserve essential positional information in words and image patches, while class tokens aggregate representative information. These strategies show promise for enhancing performance in sequence analysis [5].

In the field of biomedical signal processing, various transformer-based models have been introduced. Liu et al. [19] proposed the EEG Emotion Transformer (EeT), which modifies the traditional transformer architecture to work with EEG signals by directly learning spatial-frequency features from EEG sequences. Wang et al. [32] developed a transformer-based model for hierarchical learning, which separates spatial information from the electrode level to the brain region level, enhancing the ability to capture spatial dependencies in EEG and improving emotion recognition accuracy. Ma et al. [23] introduced a hybrid model that combines CNNs with transformers to decode motor imagery EEG signals. Li et al. [18] presented a network architecture search framework based on transformers and multi-objective evolutionary algorithms, specifically designed for emotion recognition using EEG data.

2.2. Swin Transformer

The Transformer model has achieved remarkable success in NLP and has proven competitive in image classification [34], particularly when used alongside CNNs. However, applying Transformers directly to visual tasks presents two main challenges. First, the difference in scale between images leads to large sequence lengths, causing Transformers to struggle with modeling long sequences efficiently due to their approximately quadratic scaling of parameters with respect to sequence length. Second, Transformers are less suited for dense prediction tasks, such as instance segmentation, which require fine-grained, pixel-level predictions [10]. The Swin Transformer [20] overcomes these challenges by utilizing a shifted window mechanism that reduces the number of parameters and delivers state-of-the-art performance across various pixel-level vision tasks.

In recent years, researchers have also explored the application of Swin Transformers in various EEG tasks. Wang et al. [30] integrated EEG channel attention with the Swin Transformer, marking a departure from traditional CSP and CNN models. Chen et al. [3] proposed a Spatio-Temporal Swin Transformer for emotion recognition. The incorporation of a spatio-temporal attention module enhances the model's ability to extract meaningful information from EEG signals while effectively reducing redundant noise and interference. Wei et al. [33] proposed a Transformer Capsule Network (TC-Net), which consists primarily of an EEG Transformer module for extracting EEG features and an Emotion Capsule module for refining these features and classifying emotional states.

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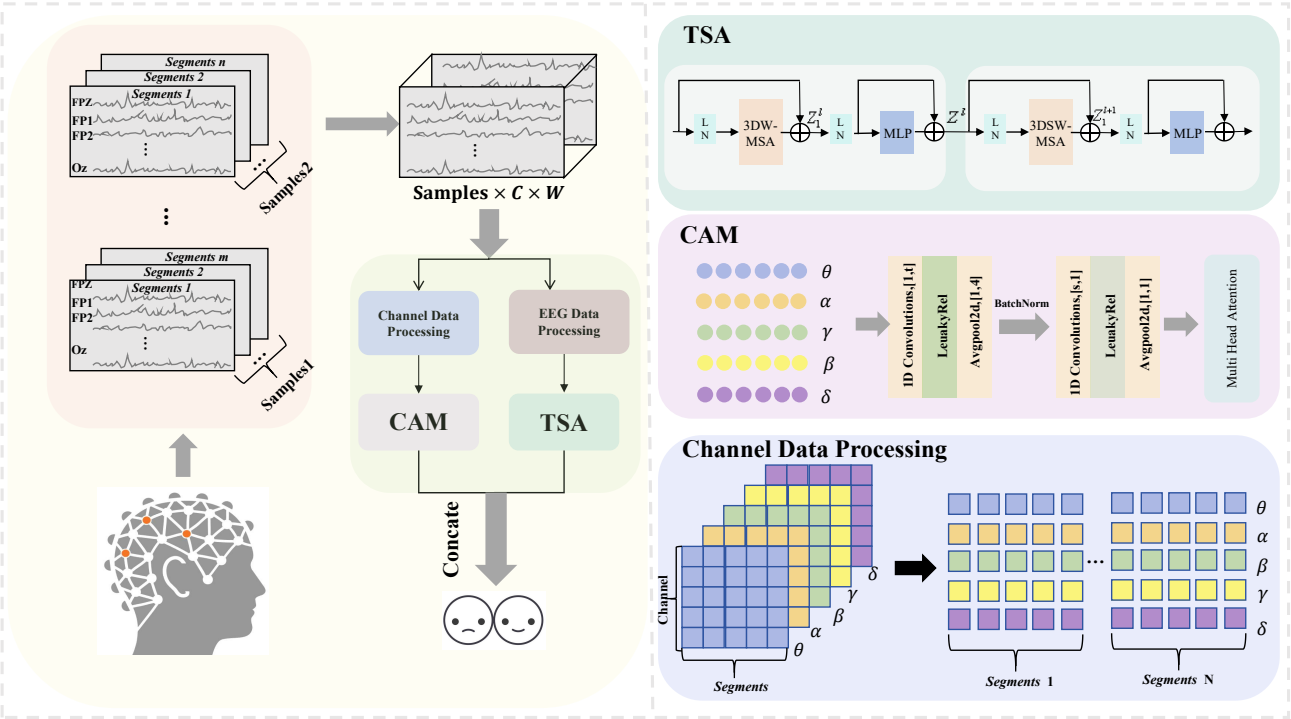

Figure 1. The overall architecture of the proposed Swin-TCNet is illustrated, where Samples denotes the number of segments, C represents the number of channels, and W indicates the window size. As shown in the figure, the detailed structure of Swin-TCNet comprises two parallel networks, each incorporating TSA and CAM modules

3. Methods

The Swin-TCNet model consists of three core components: data processing, Time-Variant Signal Attention, and channel attention module. First, the EEG signal extraction process is described, which involves sequentially connecting time windows and, for the first time, integrating EEG data along the temporal dimension. Next, the integrated data is fed into the Temporal Swin Transformer model for processing. To further extract channel-specific information from EEG signals, a channel attention mechanism is introduced. The model primarily consists of dynamic convolution layers and multi-head attention. Finally, feature-level fusion of the outputs from these two modules is performed to further enhance the model's performance. The overall framework of the model is illustrated in Figure 1.

3.1. Data Processing

The EEG signal is initially represented as $\boldsymbol{X} \in \mathbb{R}^{C\times T}$, where C denotes the number of EEG electrodes (channels), and T represents the total length of the signal.

3.1.1 EEG data processing

A discrete, non-overlapping window of length W is applied to segment the signal into blocks. Each block $\boldsymbol{X}_{\boldsymbol{C},\boldsymbol{K}} \in \mathbb{R}^{W}$ corresponds to a segment indexed by $K = 1, 2, \ldots, \lfloor \frac{T}{W} \rfloor$. Then, these blocks are stacked together to form a matrix $\boldsymbol{Z} \in \mathbb{R}^{Samples\times C\times W}$, where $Samples = \lfloor \frac{T}{W} \rfloor$ represents the total number of segments.

3.1.2 Channel Data Processing

Differential entropy (DE) was first applied in the literature for emotion recognition using EEG signals. As a metric for measuring the complexity of continuous random variables, DE is considered one of the most accurate and stable features in the field of EEG. Its mathematical expression is given as follows:

$$\begin{aligned} DE(x) &= -\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \log_2\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right) \\ &= \frac{1}{2}\log_2(2\pi e) + \frac{1}{2}\log_2(\sigma) \end{aligned} \tag{1}$$

where $x \sim N(\mu, \sigma^2)$, and π and e are constants. In this study, we analyze EEG signal features and extract DE features from five frequency bands: δ, θ, α, β, and γ, segmenting them based on time windows. Subsequently, the DE features extracted from each time segment are treated as one-dimensional features and spatially mapped according to a two-dimensional EEG electrode topology. Finally, these mapped features are used as input for the channel attention model.

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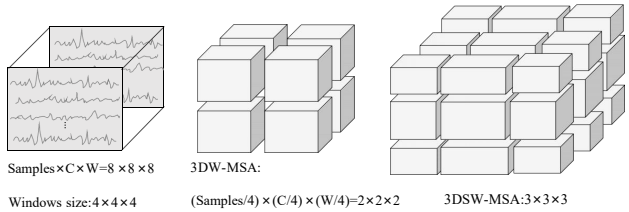

Figure 2. As shown in the figures for 3DW-MSA and 3DSW-MSA, the input size is Samples × C × W, which is 8 × 8 × 8, with a window size of 4 × 4 × 4. Due to the regular window partitioning in 3DW-MSA, the number of windows is 2 × 2 × 2 = 8. When treated as a window token in 3DSW-MSA, the number of windows increases to 27 (3 × 3 × 3)

3.2. Time-Variant Signal Attention (TSA)

The input data has a size of Samples $\times C \times W \times 1$, where Samples is the number of samples, C the channels, and W the width. A 3D convolution with a kernel and stride of $1 \times 4 \times 16$ transforms it into Samples $\times \frac{C}{4} \times \frac{W}{16}$, with each token having a 96-dimensional feature. A linear embedding layer then maps these features to an arbitrary dimension D. This operation extracts spatiotemporal features while reducing spatial and temporal resolution. The model's Time-Variant Signal Attention (TSA) framework is illustrated in Figure 2.

The Temporal Swin Transformer leverages the Transformer architecture to model global EEG dependencies, with multi-head self-attention (MSA) as its core. MSA maps input features into multiple subspaces, computes attention via dot-product operations within each, and aggregates the results back to the original space. This enables parallel extraction of diverse dependencies. The MSA mechanism is defined as follows:

$$MSA\left(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}\right)=\text{concat}\left(\boldsymbol{head}_1,\cdots,\boldsymbol{head}_{N_h}\right)\boldsymbol{W}^o \quad (2)$$

$$\boldsymbol{head}_i=\text{Attention}\left(\boldsymbol{QW}_i^Q,\boldsymbol{KW}_i^K,\boldsymbol{VW}_i^V\right) \quad (3)$$

$$\text{Attention}\left(\boldsymbol{Q}_i,\boldsymbol{K}_i,\boldsymbol{V}_i\right)=\text{Softmax}\left(\frac{\boldsymbol{Q}_i\boldsymbol{K}_i^T}{\sqrt{d_k}}\right)\boldsymbol{V}_i \quad (4)$$

Here, $\boldsymbol{head}_i$ represents the i-th attention head, where N_h denotes the total number of attention heads. Additionally, $\boldsymbol{W}^o \in \mathbb{R}^{d_v \times d_{model}}$ represents a linear mapping matrix, where d_{model} is the dimension of the input embedding.

$\boldsymbol{QW}_i^Q, \boldsymbol{KW}_i^K, \boldsymbol{VW}_i^V$ correspond to the query, key, and value for the i-th attention head, respectively. The weight matrices are defined as follows: $\boldsymbol{W}_i^Q \in \mathbb{R}^{d_{model}\times d_q}, \boldsymbol{W}_i^K \in \mathbb{R}^{d_{model}\times d_k}, \boldsymbol{W}_i^V \in \mathbb{R}^{d_{model}\times d_v}$, where d_q, d_k, d_v denote the dimensions of the query, key, and value, respectively.

In the temporal Swin block, a standard temporal Swin block is introduced to extract semantic features from three-dimensional frames. Specifically, the temporal Swin block consists of two consecutive components: a Three-Dimensional Window Multi-Head Self-Attention (3DW-MSA) mechanism and a Three-Dimensional Shifted Window Multi-Head Self-Attention (3DSW-MSA) mechanism.

Before performing 3DW-MSA and 3DSW-MSA computations, the input sequence first undergoes layer normalization. The results of 3DW-MSA and 3DSW-MSA are subsequently processed through residual connections and layer normalization before being fed into a multilayer perceptron (MLP). The formulas for 3DW-MSA and 3DSW-MSA are as follows:

$$\mathrm{z}_1^1=3DW\text{-}MSA\left(LN\left(\mathrm{z}^{1-1}\right)\right)+\mathrm{z}^{1-1} \quad (5)$$

$$z^l=FFN\left(LN\left(\mathrm{z}_1^1\right)\right)+\mathrm{z}_1^1 \quad (6)$$

$$\mathrm{z}_1^{1+1}=3DSW\text{-}MSA\left(LN\left(z^l\right)\right)+z^l \quad (7)$$

$$z^{l+1}=FFN\left(LN\left(\mathrm{z}_1^{1+1}\right)\right)+\mathrm{z}_1^{1+1} \quad (8)$$

The variables $\boldsymbol{z}_1^l$ and $\boldsymbol{z}^l$ represent the output features of the 3D (S)W-MSA module and the FFN module, respectively. The 3DW-MSA and 3DSW-MSA refer to the multi-head self-attention mechanisms based on 3D windows, which use regular and shifted window partition configurations, respectively.

The features extracted from the Temporal Swin Transformer are represented as $R^{D\times Samples\times\frac{C}{4}\times\frac{W}{16}}$, where D denotes the number of hidden layer channels in the Temporal Swin Transformer, while *Samples*, $\frac{C}{4}$, and $\frac{W}{16}$ correspond to temporal frames, height, and width, respectively.

3.3. Channel Attention Module (CAM)

To enhance the model's focus on channel information, a Channel Attention Module (CAM) is introduced. The model integrates dynamic temporal and spatial layers with a multi-head attention layer. Given EEG signals' strong temporal characteristics, 1D convolution effectively captures dynamic variations along the temporal and channel axes, improving feature extraction and computational efficiency. Thus, 1D convolutional kernels are used in this layer alongside the Transformer to reinforce CAM.

3.3.1 Dynamic Time Convolution

In the dynamic temporal convolution layer, to enable the neural network to learn dynamic temporal representations, let X represent a 2D DE feature sample of EEG signals, expressed as $\mathrm{X}=\left[\mathrm{X}^0,\mathrm{X}^1,\ldots,\mathrm{X}^n\right]$, where $\mathrm{X}^n \in \mathbb{R}^{c\times l}$. The variable n represents the total number of DE feature samples in a single experiment, while c denotes the number of frequency bands. Additionally, l represents the number of DE feature samples for all channels within each time period. Dynamic features are obtained by applying different types of convolution kernels to these samples. Subsequently, $LeakyReLU\left(\cdot\right)$ is used as the feature activation function, followed by average pooling to downsample the feature maps, reducing noise and mitigating the dimensional effects of the features. Let Z_i be the output of the i-th type of convolution in the dynamic temporal layer; then we have $Z_i \in R^{n\times m\times c\times l_i}$, where n is the number of samples, m is the number of output feature channels from the convolution, c represents the number of channels, and l_i denotes the feature length of the i-th convolution. It is defined as follows:

$$Z_i{=}AvgPool\left(LeakyReLU\left(Conv\left(X,S_t^1\right)\right)\right) \quad (9)$$

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where the multi-scale $1D$ convolution kernel $(1,t)$ enriches the learning of temporal information and is denoted as S_t^1, where $t \in [16, 32, 64]$. $LeakyReLU\,(\cdot)$ is the activation function. Additionally, $AvgPool\,(\cdot)$ denotes the average pooling operation. Each output is concatenated along the feature dimension.

To address the issue of internal covariate shift in neural networks, a normalization operation is applied after the dynamic temporal convolution layer. Thus, the final output of this layer is represented as $\mathrm{Z_T}$, where $\mathrm{Z_T} \in \mathrm{R}^{\mathrm{n\times m\times c\times \int l_i}}$, defined as follows:

$$Z_T{=}f_{bn}\left([Z_1, \ldots, Z_i]\right), i \in [1, 2, 3] \tag{10}$$

where f_{bn} denotes the normalization operation, and Z_i represents the concatenation of the outputs of the i-th type of convolution along the feature dimension.

3.3.2 Dynamic Spatial Convolution

In this dynamic spatial convolution layer, let Z_j be the output of the j-th dynamic temporal convolution in this layer; then we have $Z_j \in R^{n\times m\times c_j\times l_j}$, where n represents the sample size, m denotes the number of output feature channels after convolution, c_j represents the number of channels after the j-th convolution operation, and l_j indicates the feature length after the j-th convolution. The definition of Z_j is as follows:

$$Z_j{=}AvgPool\left(LeakyReLU\left(Conv\left(Z_T, S_1^S\right)\right)\right) \tag{11}$$

A multi-scale one-dimensional convolution kernel $(\mathrm{s}, 1)$ is used to learn rich spatial information, where the size of s is related to the number of EEG signal channels. This layer employs two types of convolution kernels: global convolution kernels and local convolution kernels. In the global convolution kernel, s corresponds to the number of channels, specifically set to 5. Since the length of the convolution kernel matches the channel dimension, it effectively learns global spatial information. In contrast, in the local convolution kernel, s is set to 2, allowing the extraction of cross-channel spatial information, denoted as S_1^S, where $S \in [5, 2]$. The activation function used is $LeakyReLU\,(\cdot)$, while $AvgPool\,(\cdot)$ represents the average pooling operation. Each output is concatenated along the channel dimension.

Similar to the dynamic temporal convolution layer, the spatial convolution layer also requires a normalization operation; therefore, the final output of this layer is defined as Z_p, where $Z_p \in \mathrm{R}^{\mathrm{n\times m\times \int c_i\times l_i}}$, and is defined as follows:

$$Z_p{=}f_{bn}\left([Z_1, \ldots, Z_j]\right), j \in [1, 2] \tag{12}$$

where f_{bn} denotes the normalization operation, and Z_p represents the concatenation of the outputs of the j-th convolution type along the feature dimension.

3.3.3 Multi Head Attention

In the multi-head attention layer, the output data from the dynamic convolution layer is received and undergoes dimensional transformation, resulting in the dataset dimension represented as $Z_{1\mathrm{D}} \in R^{n\times c\times l}$, where n is the number of samples, c is the number of frequency bands after the dynamic convolution layer, and l is the number of features after the dynamic convolution layer. First, the input data undergoes a linear mapping to compute the Query (Q), Key (K), and Value (V):

$$Q = XW_{\mathrm{q}}\,, \mathrm{K} = \mathrm{X}W_{\mathrm{k}}\,, \mathrm{V} = \mathrm{X}W_{\mathrm{v}} \tag{13}$$

where $\mathrm{W_q}$, $\mathrm{W_k}$, and $\mathrm{W_v}$ are learnable weight matrices. Then, Q, K, and V are split into h attention heads, with each head having a feature dimension of $d_h = \frac{d}{h}$. Each head independently computes self-attention, and the outputs of multiple attention heads are concatenated and then linearly transformed. The specific formulas are as follows:

$$\mathrm{Attention(Q_i, K_i, V_i)} = SoftMax\left(\frac{QK^T}{\sqrt{d}}\right)x^l \tag{14}$$

$$\mathrm{MultiHead(Q, K, V) = Concat(head_1, ..., head_n)W_o} \tag{15}$$

where $\mathrm{W_o}{\in}R^{\mathrm{d_{model}\times d_{model}}}$ is the final learnable weight matrix. The final output dimension is $R^{n\times c\times \mathrm{d_{model}}}$.

The specific steps of the channel attention layer are summarized as follows:

(1) A $(1,t)$ convolution kernel is used to perform convolution operations on the two-dimensional dynamic features.

(2) The output obtained in the first step is concatenated along the feature dimension to obtain dynamic temporal features.

(3) A $(s, 1)$ convolution kernel is applied to the output of the second step for further convolution computation.

(4) The output of the third step is concatenated along the channel dimension to extract cross-channel spatial features.

(5) Multi-head attention is used to fuse the processed cross-channel spatiotemporal features, thereby integrating two types of channel feature information.

3.4. Classification

The feature representations of the above two modules are denoted as z^{l_v} and z^{l_c}, are concatenated at the feature level. The fused features are then passed through a multilayer perceptron (MLP) for classification via a fully connected layer. The MLP output layer has neurons corresponding to the number of categories. The Softmax function is applied to compute predicted probabilities, with the highest probability determining the final classification. Cross-entropy loss is used as the loss function for Swin-TCNet. The formulas are as follows: In this method, we first perform feature concatenation: $z^l = \mathrm{Concat}(z^{l_v}, z^{l_c})$, then apply an MLP transformation: $z^{\mathrm{mlp}} = \varphi(\gamma(z^l))$, followed by Softmax to compute probabilities: $P = \mathrm{Softmax}(z^{\mathrm{mlp}})$, and finally, we compute the cross-entropy loss: $\tau = -\sum_{i=1}^{C} y_i \log P_i$. where $\varphi(\cdot)$ is the activation function GeLU, $\gamma(\cdot)$ is a linear layer, y_i represents the ground-truth label, P_i represents the predicted label, and C is the number of categories. Through feature-level fusion, the MLP classifier, and cross-entropy loss, a complete classification pipeline is constructed.

4. Experiments

4.1. Dataset

The effectiveness of the proposed Swin-TCNet model was evaluated on two publicly available EEG datasets, SEED and SEED-IV. The details are as follows:

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Table 1. Subject-independent emotion recognition accuracy (mean ± standard deviation) in the SEED and SEED-IV datasets

Model	SEED	SEED-IV
RGNN[39]	85.30 ± 6.72	73.84 ± 8.02
SOGNN[15]	86.81 ± 5.79	75.27 ± 8.19
MR-VAE[27]	92.83 ± 6.04	79.30 ± 7.53
MSS-JDA[2]	93.78 ± 3.39	78.93 ± 7.39
SFT-SGAT[26]	92.04 ± 3.28	82.76 ± 4.02
FC-FAN[7]	92.12 ± 3.32	78.98 ± 5.17
Our Method (PTSwin-Net)	**93.47 ± 3.69**	**86.80 ± 9.68**

SEED[37]: This dataset consists of EEG signals recorded from 15 Chinese participants (7 males and 8 females) using 62 channels. The participants watched 15 movie clips designed to elicit three types of emotions: positive, negative, and neutral. Each movie clip corresponds to an experiment, and each participant watched the same set of clips three times at different sessions.

SEED-IV[36]: This multimodal dataset includes EEG and eye-tracking data collected from 15 right-handed healthy participants (7 males and 8 females). A total of 45 experiments were conducted across three sessions, with each session featuring different stimulus materials. During each session, participants watched 24 movie clips designed to evoke four emotions: happiness, sadness, fear, and neutrality.

We used precomputed EEG features from the SEED and SEED-IV datasets. Raw signals were downsampled from 1000 Hz to 200 Hz, and differential entropy (DE) features were extracted via Short-Time Fourier Transform (STFT). SEED used a 1-second window, while SEED-IV used a 4-second window, both without overlap, across five frequency bands: δ (1–4 Hz), θ (4–8 Hz), α (8–14 Hz), β (14–31 Hz), and γ (31–50 Hz).

Table 2. SEED and SEED-IV subject classification accuracy

Subject	Accuracy (%)	
	SEED	SEED-IV
1	93.60	88.67
2	87.61	90.89
3	91.60	62.22
4	95.60	90.89
5	97.61	78.67
6	93.60	78.33
7	97.61	70.88
8	95.61	95.33
9	97.61	93.11
10	87.61	86.44
11	97.61	96.44
12	93.60	90.89
13	89.60	92.11
14	95.61	97.56
15	87.61	89.56
Average	93.47	86.80

4.2. Implementation Details

All experiments were performed on an NVIDIA GeForce RTX 3090 GPU. The model was trained for 500 epochs using the Adam optimizer with a batch size of 16. The initial learning rate was set to 0.0001 and adjusted via a cosine annealing schedule for stable convergence. Key hyperparameters were selected through grid search based on validation performance. The Temporal Swin Transformer module employed six layers of 3D window-based multi-head self-attention (3DW-MSA) with a window size of $16 \times 7 \times 7$, balancing temporal modeling and efficiency. The dynamic temporal convolutional layer used multi-scale kernels of size $(1, t)$, where $t \in \{16, 32, 64\}$, to capture temporal patterns at different scales. Similarly, the spatial convolutional layer adopted kernel sizes of $(s, 1)$, where $s \in \{5, 2\}$, for multi-resolution spatial feature extraction. These settings consistently achieved optimal validation accuracy across datasets.

To evaluate the performance of Swin-TCNet on the SEED and SEED-IV datasets, this study conducted cross-subject experiments to verify the effectiveness of Swin-TCNet in EEG-based emotion recognition. For a fair comparison with previous studies, we adopted the Leave-One-Subject-Out (LOSO) cross-validation strategy on both datasets. At each iteration, the samples from one subject were used as the test set, while the remaining subjects' samples were combined to form the training set.

4.3. Ablation Study

Ablation experiments on the SEED and SEED-IV datasets were conducted to assess each module's contribution to Swin-TCNet's performance. Results, presented in tabular format, highlight the impact of key modules, including time-varying signal attention and channel attention.

Table 3. The ablation study of CAM and TSA on the SEED and SEED-IV datasets (Accuracy in %)

Dataset	CAM	TSA	Accuracy (%)
SEED	✓	×	85.26 ± 6.03
	×	✓	92.37 ± 4.57
	✓	✓	93.47 ± 3.69
SEED-IV	✓	×	81.33 ± 9.03
	×	✓	85.04 ± 8.09
	✓	✓	86.80 ± 9.68

We evaluated each module's contribution via ablation studies (Table 3). The channel attention module enhanced inter-channel interactions, achieving 85.26% and 81.33% accuracy on SEED and SEED-IV. Incorporating the Temporal Swin Transformer, which captures temporal dependencies and enables hierarchical feature extraction, improved accuracy to 92.37% and 85.04%. Its window attention and shifted window strategy further boost local and global EEG feature modeling, enhancing classification.

To refine temporal modeling, we conducted ablation studies on temporal and 3D shifting within the TSA module. As shown

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Table 4. Ablation study of temporal shifting and 3D shifting in TSA on SEED and SEED-IV datasets

Dataset	temporal shifting	3D shifting	Accuracy (%)
SEED	×	×	82.26 ± 9.23
	✓	×	90.36 ± 5.87
	✓	✓	92.37 ± 4.57
SEED-IV	×	×	76.33 ± 9.78
	✓	×	82.32 ± 9.09
	✓	✓	85.04 ± 8.09

Table 5. Ablation study on the temporal window size of 3D tokens

Dataset	Window Size	Accuracy (%)
SEED	4×7×7	91.21 ± 4.89
	8×7×7	91.68 ± 4.61
	16×7×7	92.37 ± 4.57
SEED-IV	4×7×7	84.89 ± 8.16
	8×7×7	84.91 ± 8.29
	16×7×7	85.04 ± 8.09

in Table 4, temporal shifting alone significantly improved accuracy, while its combination with 3D shifting further boosted performance to 92.37% on SEED and 85.04% on SEED-IV. These results demonstrate that temporal shifting effectively captures dynamic EEG patterns, and 3D shifting enhances spatiotemporal representations.

We also examined the effect of temporal window size in 3D tokenization. As shown in Table 5, larger temporal windows consistently improved accuracy—on SEED, from 91.21% (4×7×7) to 92.37% (16×7×7), and on SEED-IV, from 84.89% to 85.04%. This indicates that broader temporal context facilitates better long-range dependency modeling, enhancing overall performance.

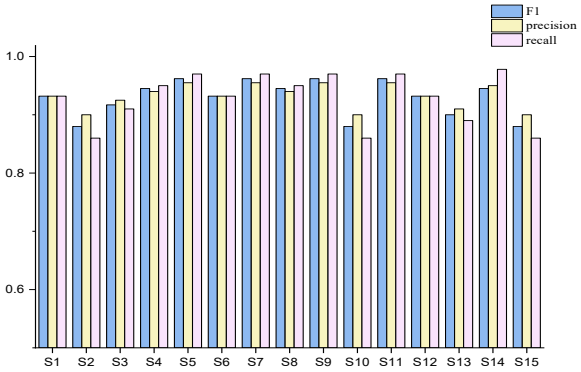
4.4. Comparative Study

To validate the effectiveness of the Swin-TCNet network, we compared its classification performance with that of major mainstream networks on the SEED and SEED-IV datasets. Table 1 presents the subject-independent emotion recognition accuracy of each model on both datasets.

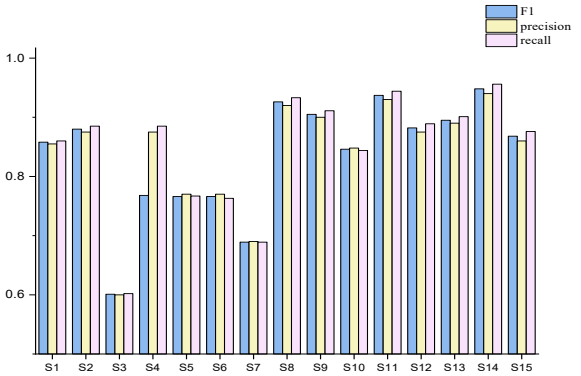
Compared to other mainstream networks, the proposed Swin-TCNet network achieves the best performance on both datasets, with an average accuracy of 93.47% on the SEED dataset and 86.80% on the SEED-IV dataset.

4.5. Evaluation Metrics

Subjects are indexed from S1 to S15. Figures 3a and 3b present the emotion classification results on the SEED and SEED-IV datasets. SEED comprises three emotion categories (Sad, Neutral, Happy), while SEED-IV includes four (Happy, Sad, Neutral, Fear). On SEED, the accuracy, F1-score, recall, and precision reach 93.47%, 93.49%, 93.50%, and 93.52%, respectively.



(a) SEED dataset



(b) SEED-IV dataset

Figure 3. F1 score, precision, and recall of Swin-TCNet for each individual.

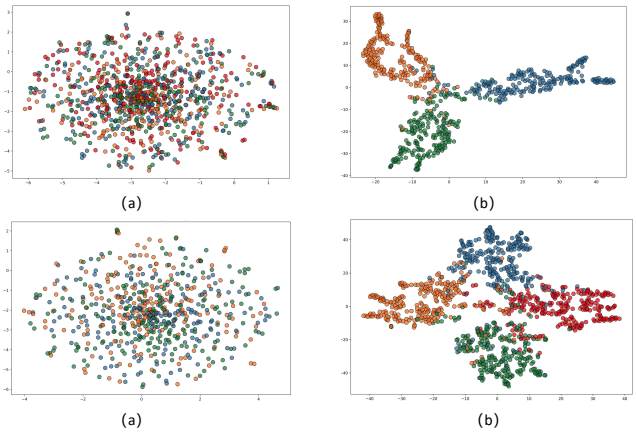


Figure 4. The t-SNE visualizations for independent experiments using the SEED and SEED-IV datasets. Different colors represent different emotions: green indicates sad emotion, orange represents neutral emotion, blue denotes happy emotion, and red indicates fear. (a) Initial data distribution; (b) Integrating the data distribution of two modules

For SEED-IV, these metrics are 86.76%, 86.70%, 86.70%, and 86.70%.

The confusion matrices in Figures 5a and 5b illustrate the recognition results of Swin-TCNet corresponding to the best-

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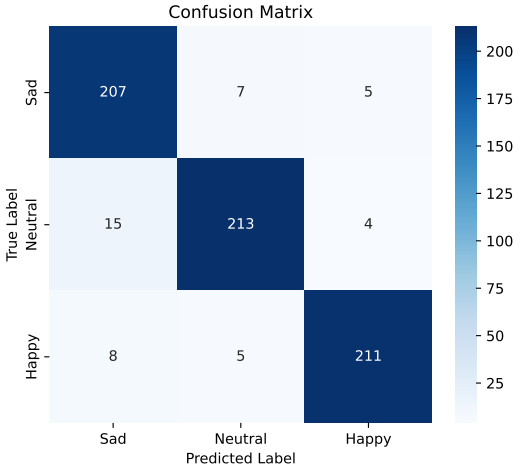
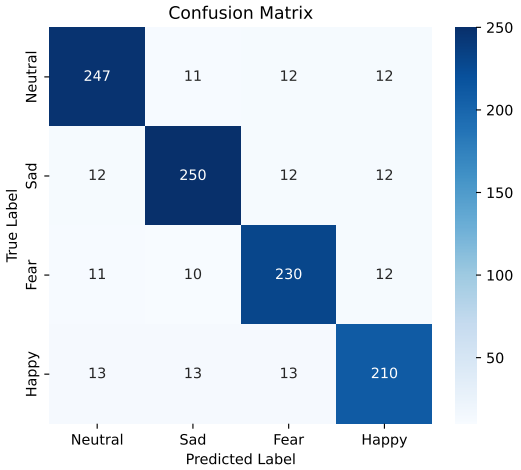
performing models. The horizontal and vertical axes denote predicted and true labels, respectively. On SEED, the neutral emotion is more distinctly recognized, while on SEED-IV, neutral and fear exhibit higher recognition accuracy compared to sadness and happiness.

Subjects S3 and S7 show consistently lower performance across both datasets, likely due to individual differences in emotional expression or lower EEG signal quality. Their feature embeddings exhibit higher intra-class variance and inter-class overlap, making it harder for the model to form clear decision boundaries.

To further investigate the model's learning behavior, Subject S10—who demonstrated stable and representative performance close to the overall average—was selected for visualization (Figure 4). The t-SNE plots depict Swin-TCNet's feature representations: the first row corresponds to SEED and the second to SEED-IV. Stage (a) shows the initial distribution, while stage (b) illustrates the fused features from two modules. As training progresses, feature clusters become more distinct, confirming the model's effectiveness in improving class separability.



(a) SEED datasets



(b) SEED-IV datasets

Figure 5. Confusion matrices for SEED and SEED-IV datasets.

5. Conclusion

This paper introduces the Swin-TCNet model, specifically designed for emotion recognition tasks. The model incorporates a channel attention module and a Temporal Swin Transformer module, enabling the dynamic extraction of temporal dependencies across different time intervals. Through this mechanism, it generates highly discriminative fused features. Ultimately, the model aggregates information from diverse temporal segments to construct a novel feature representation. This approach not only effectively accounts for individual variability but also significantly enhances the model's generalization capability. Comprehensive qualitative and quantitative evaluations on multiple benchmark datasets demonstrate that Swin-TCNet surpasses state-of-the-art methods, highlighting its superior effectiveness. Beyond its outstanding performance in emotion recognition tasks, the model also exhibits strong potential for applications in related domains. By fully leveraging temporal dynamics and channel-specific feature representations, Swin-TCNet provides a more precise and holistic analysis for emotion recognition. Future research could explore the integration of this approach with multimodal emotion recognition techniques to further improve classification performance.

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References

[1] Z. Cai, Y. Gao, F. Fang, Y. Zhang, and S. Du. Multi-layer transfer learning algorithm based on improved common spatial pattern for brain–computer interfaces. *Journal of Neuroscience Methods*, 415:110332, 2025.

[2] S. Chen, W. Ma, Y. Wang, and X. Sun. Mss-jda: Multi-source self-selected joint domain adaptation method based on cross-subject eeg emotion recognition. *Biomedical Signal Processing and Control*, 100:106953, 2025.

[3] Z. Chen, J. Jin, and J. Pan. Spatio-temporal swin transformer-based 4-d eeg emotion recognition. In *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 1850–1855. IEEE, 2023.

[4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.

[5] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

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[6] S. Garg, R. K. Patro, S. Behera, N. P. Tigga, and R. Pandey. An overlapping sliding window and combined features based emotion recognition system for eeg signals. *Applied Computing and Informatics*, 21(1/2):114–130, 2025.

[7] W. Guo, Y. Li, M. Liu, R. Ma, and Y. Wang. Functional connectivity-enhanced feature-grouped attention network for cross-subject eeg emotion recognition. *Knowledge-Based Systems*, 283:111199, 2024.

[8] N. Hollenstein, C. Renggli, B. Glaus, M. Barrett, M. Troendle, N. Langer, and C. Zhang. Decoding eeg brain activity for multi-modal natural language processing. *Frontiers in Human Neuroscience*, 15:659410, 2021.

[9] E. H. Houssein, A. Hammad, and A. A. Ali. Human emotion recognition from eeg-based brain–computer interface using machine learning: a comprehensive review. *Neural Computing and Applications*, 34(15):12527–12557, 2022.

[10] B. Hu, Z. Ye, Z. Wei, E. Snezhko, V. Kovalev, and M. Ye. Mlda-net: Multi-level deep aggregation network for 3d nuclei instance segmentation. *IEEE Journal of Biomedical and Health Informatics*, 2025.

[11] X. Hu, J. Chen, F. Wang, and D. Zhang. Ten challenges for eeg-based affective computing. *Brain Science Advances*, 5(1):1–20, 2019.

[12] W. Huang, W. Wang, Y. Li, and W. Wu. Fbstcnet: A spatio-temporal convolutional network integrating power and connectivity features for eeg-based emotion decoding. *IEEE Transactions on Affective Computing*, 2024.

[13] W.-B. Jiang, X.-H. Liu, W.-L. Zheng, and B.-L. Lu. Seed-vii: A multimodal dataset of six basic emotions with continuous labels for emotion recognition. *IEEE Transactions on Affective Computing*, 2024.

[14] R. Joshi and M. Jadeja. The synergy of clinical psychology and affective computing: Advancements in emotion recognition and therapy. In *Affective Computing for Social Good: Enhancing Well-being, Empathy, and Equity*, pages 21–45. Springer, 2024.

[15] J. Li, S. Li, J. Pan, and F. Wang. Cross-subject eeg emotion recognition with self-organized graph neural network. *Frontiers in Neuroscience*, 15, June 2021.

[16] J. Li, S. Qiu, C. Du, Y. Wang, and H. He. Domain adaptation for eeg emotion recognition based on latent representation similarity. *IEEE Transactions on Cognitive and Developmental Systems*, 12(2):344–353, 2019.

[17] J. Li, S. Qiu, Y.-Y. Shen, C.-L. Liu, and H. He. Multisource transfer learning for cross-subject eeg emotion recognition. *IEEE transactions on cybernetics*, 50(7):3281–3293, 2019.

[18] X. Li, Y. Zhang, P. Tiwari, D. Song, B. Hu, M. Yang, Z. Zhao, N. Kumar, and P. Marttinen. Eeg based emotion recognition: A tutorial and review. *ACM Computing Surveys*, 55(4):1–57, 2022.

[19] J. Liu, H. Wu, L. Zhang, and Y. Zhao. Spatial-temporal transformers for eeg emotion recognition. In *Proceedings of the 6th International Conference on Advances in Artificial Intelligence*, pages 116–120, 2022.

[20] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 10012–10022, 2021.

[21] W. Lu, T.-P. Tan, and H. Ma. Bi-branch vision transformer network for eeg emotion recognition. *IEEE Access*, 11:36233–36243, 2023.

[22] Y. Ma, Z. Huang, Y. Yang, Z. Chen, Q. Dong, S. Zhang, and Y. Li. Msbilstm-attention: Eeg emotion recognition model based on spatiotemporal feature fusion. *Biomimetics*, 10(3):178, 2025.

[23] Y. Ma, Y. Song, and F. Gao. A novel hybrid cnn-transformer model for eeg motor imagery classification. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE, 2022.

[24] S. Pal, S. Mukhopadhyay, and N. Suryadevara. Development and progress in sensors and technologies for human emotion recognition. *Sensors*, 21(16):5554, 2021.

[25] V. Pandey, N. Panwar, A. Kumbhar, P. P. Roy, and M. Iwamura. Enhanced cross-task eeg classification: Domain adaptation with eegnet. In *International Conference on Pattern Recognition*, pages 354–369. Springer, 2025.

[26] L. Qiu, L. Zhong, J. Li, W. Feng, C. Zhou, and J. Pan. Sft-sgat: A semi-supervised fine-tuning self-supervised graph attention network for emotion recognition and consciousness detection. *Neural Networks*, 180:106643, 2024.

[27] J. Quan, Y. Li, L. Wang, R. He, S. Yang, and L. Guo. Eeg-based cross-subject emotion recognition using multi-source domain transfer learning. *Biomedical Signal Processing and Control*, 84:104741, 2023.

[28] W. Tao, C. Li, R. Song, J. Cheng, Y. Liu, F. Wan, and X. Chen. Eeg-based emotion recognition via channel-wise attention and self attention. *IEEE Transactions on Affective Computing*, 14(1):382–393, 2020.

[29] A. Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

[30] H. Wang, L. Cao, C. Huang, J. Jia, Y. Dong, C. Fan, and V. H. C. De Albuquerque. A novel algorithmic structure of eeg channel attention combined with swin transformer for motor patterns classification. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2023.

[31] Y. Wang, B. Zhang, and L. Di. Research progress of eeg-based emotion recognition: A survey. *ACM Computing Surveys*, 56(11):1–49, 2024.

[32] Z. Wang, Y. Wang, C. Hu, Z. Yin, and Y. Song. Transformers for eeg-based emotion recognition: A hierarchical spatial information learning model. *IEEE Sensors Journal*, 22(5):4359–4368, 2022.

[33] Y. Wei, Y. Liu, C. Li, J. Cheng, R. Song, and X. Chen. Tc-net: A transformer capsule network for eeg-based emotion recognition. *Computers in biology and medicine*, 152:106463, 2023.

[34] L. Yuan, Y. Chen, T. Wang, W. Yu, Y. Shi, Z.-H. Jiang, F. E. Tay, J. Feng, and S. Yan. Tokens-to-token vit: Training vision transformers from scratch on imagenet. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 558–567, 2021.

[35] G. Zhao, Y. Ge, B. Shen, X. Wei, and H. Wang. Emotion analysis for personality inference from eeg signals. *IEEE transactions on affective computing*, 9(3):362–371, 2017.

[36] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki. Emotionmeter: A multimodal framework for recognizing human emotions. *IEEE transactions on cybernetics*,

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49(3):1110–1122, 2018.

[37] W.-L. Zheng and B.-L. Lu. Investigating critical frequency bands and channels for eeg-based emotion recognition with deep neural networks. *IEEE Transactions on autonomous mental development*, 7(3):162–175, 2015.

[38] W.-L. Zheng, J.-Y. Zhu, and B.-L. Lu. Identifying stable patterns over time for emotion recognition from eeg. *IEEE transactions on affective computing*, 10(3):417–429, 2017.

[39] P. Zhong, D. Wang, and C. Miao. Eeg-based emotion recognition using regularized graph neural networks. *IEEE Transactions on Affective Computing*, 13(3):1290–1301, July 2022.

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